Localization by Fine-Grained Map Retrieval: Geo-Localization of Remote Sensing Images Using Vector Map Tiles

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***Abstract*—Accurate geo-localization of remote sensing images without explicit geographic metadata remains a persistent challenge in the remote sensing field. Most existing approaches rely on matching query images against prereferenced imagery; however, these often struggle with dynamic field changes, viewpoint variations, domain gaps, and scale inconsistencies. In contrast, vector maps, which have already undergone feature representation, selection, and generalization, can reduce redundant information and thus offer a valuable alternative. Inspired by map-based registration methods, we propose a novel framework that geo-localizes remote sensing images by retrieving map tiles containing only linear symbols at fine granularity. First, vector data are cartographically converted into multiscale linear symbol map tiles embedding precise location information. This approach helps minimize overlap and reduce map size. Next, a dual-stage attention multilevel feature model (DSA-MLM) uses the query remote sensing image to retrieve the most relevant map tiles. Finally, a cluster-based geo-localization strategy estimates the potential geographic location of the query image. Experimental results across various geographic regions, sensor types, resolutions, and map conditions demonstrate that the proposed method achieves high retrieval and geo-localization accuracy, as well as efficiency and robustness in real-world scenarios, highlighting its generalizability and reliability.**

***Index Terms*—Fine-grained retrieval, geo-localization, remote sensing images, vector map.**

I. Introduction

DUE to the discontinuity of global navigation satellite system (GNSS) signals, data acquisition errors, or incomplete geographic information of the coverage area, remote sensing images, especially that are taken by uncrewed aerial vehicle (UAV), often lack spatial reference information. Consequently, supplementing the missing spatial reference information, naming geo-localization, plays a critical role in the following remote sensing application. Geo-localization of remote sensing images, including image registration, has already found widespread applications in various fields, including cargo transport [1] and UAV navigation [2]. One popular way of remote sensing image geo-localization is through retrieval, where the most similar reference files containing location information (i.e., with geographic landmarks or coordinates) are retrieved from a database, thereby providing the location information corresponding to the query image [3], [4], [5], [6].

Existing geo-localization methods of remote sensing images can be classified into two categories by their reference file type: image-based and map-based. Image-based geo-localization typically relies on the matching of query images with well-referenced remote sensing images [7], [8], [9], [10], [11], [12]. Traditional methods based on handcrafted features have been used to address this issue [1], [13], [14], [15], such as SIFT [16] and SURF [17]. With breakthroughs in deep learning technology in computer vision, an increasing number of studies have shifted toward using deep neural networks [18], [19], [20], such as convolutional neural networks (CNNs), to learn the matching relationships between query images and satellite images. Despite this, the image-based matching still faces challenges such as field changes, viewpoint differences, domain gap, and scale inconsistencies. Additionally, image-based methods are often affected by factors such as redundant information like seasonal differences, which further limit the matching effectiveness and efficiency [11], [21].

To address these challenges, some researchers have turned to map-based geo-localization methods, which uses existing maps as a reference. Since the mapping process may already include feature representation, semantic information, and feature generalization, it may effectively eliminate the low-level information and thus provide a more stable and simple representation than the reference of remote sensing image. Some maps are directly generated from remote sensing images using deep learning techniques through semantic segmentation [22] or object detection [23], [24]. Specifically, some methods establish feature correspondences between road intersections and remotely sensed semantic maps [25], [26], [27], [28]. However, these methods either require manual identification of intersections or depend on image sequences to capture key features. Because they rely on handcrafted feature priors, their generalization capability remains questionable. Moreover, after the semantic map is created, matching is still performed within the linear-feature modality rather than through a truly cross-modal approach. Other methods perform cross-modal

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matching between images and map tiles in small regions, such as utilizing OpenStreetMap (OSM) for map generation or extracting building contours and road network information from databases like Digimap and Ordnance Survey (OS) Mastermap for geo-localization tasks [29], [30]. While these methods are less affected by environmental and weather conditions, the maps that they generated often contain comprehensive mapping rules of different feature types, making it time-consuming to create the map tiles and difficult for the matching algorithm to extract features like long straight lines from cluttered backgrounds, which complicates the matching process.

In response to these challenges, we propose a geo-localization method for remote sensing imagery based on fine-grained retrieval of map tiles with only linear symbols. Specifically, vector spatial data are first converted into map tiles, which are then used as candidate reference files and matched with query remote sensing images through fine-grained retrieval by a dual-stage attention multilevel feature model (DSA-MLM). The cartography process employed in this process uses only linear symbols like the boundaries of building. These are mainly due to the ubiquity of linear symbols in maps. Also, using only linear symbols may reduce computational complexity, reduce the occlusion among features, simplify cross-domain matching, and thus reduce matching errors caused by factors like viewpoint differences, enhancing the robustness of the matching process. The DSA-MLM first extracts and aligns salient features from remote sensing images and map tiles. Then, it calculates the feature similarity between the query image and map tiles for similarity-based fine-grained map tile retrieval. Based on the retrieval results, we employ a cluster-based geo-localization strategy to filter the optimal map tile group. Subsequently, a bivariate normal distribution is utilized to statistically compute the confidence interval, thereby determining the geographic boundary of the query image. Extensive experiments demonstrate that the method achieves superior geo-localization accuracy across diverse geographic regions and image resolutions, confirming its robustness, generalization, and reliability in practical applications.

In summary, the main contributions of this article are as follows.

1) We proposed a novel geo-localization by map retrieving method for remote sensing imagery using map tiles with only linear symbol as reference. This method integrates map with cross-modal deep learning content-based retrieval technique. Compared to existing geo-localization methods for remote sensing imagery, our approach simplifies the data preparation, significantly reduces localization error, improves query efficiency, and enables localization in a large research region.
2) We designed a novel DSA-MLM for fine-grained cross-modal retrieval between remote sensing image and map tiles with linear symbols. By employing a dual-stage attention mechanism strategy and deep similarity quantification, this model can extract features at different scales and perform cross-modal retrieval with outstanding accuracy and robustness.
3) We invented an innovative cluster-based geo-localization strategy based on the results of fine-grained cross-modal retrieval. The method clusters retrieved map tiles based on centroid distances within a predefined radius threshold and selects the optimal group by cardinality for geo-localization. The 50% confidence interval from the center point of this group, which is calculated using Mahalanobis distance, defines the query image's location result.

The structure of this article is organized as follows. Section II provides an overview of related methods in the fields of geo-localization and cross-modal retrieval. Section III elaborates the proposed methodology. Section IV presents the experimental process and provides a comparative analysis. Section V concludes this article.

II. Related Work

A. Geo-Localization of Remote Sensing Imagery

Geo-localization of remote sensing images aims to determine the geographic location of an image that lacks reference information. Common practices match the query image to geo-referenced files, such as aligning remote sensing images with geo-tagged map tiles for localization, so the locational information of the matched files can be transferred to the query image. These algorithms often face cross-view, cross-domain, and cross-modal challenges in matching, as the query image and the candidate files may have different viewpoints, using different sensors, acquired in different imaging conditions, and even in different data forms.

Cross-view retrieval refers to the alignment and matching of candidate images from different viewpoints, with particular challenges in aligning ground and aerial views. The common goal of these methods is to optimize matching and registration between images from different viewpoints, overcoming the challenges posed by viewpoint variation [31], [32], [33]. Rodrigues and Tani [4] proposed a geo-localization method through retrieving corresponding aerial views from large geo-tagged aerial image databases. Hu et al. [5] introduced a deep learning cross-view matching network for geo-localization tasks from ground views to aerial views. To improve the matching accuracy between ground and satellite images, researchers have adopted various methods, such as UAV perspective technology [34], [35], [36], [37], transformer-based feature representation learning with heatmap distribution analysis [7], and local-to-global feature aggregation techniques [32], [38]. These approaches enhance feature understanding, capture global contextual information, and integrate local and global features, significantly improving the accuracy and robustness of cross-view geo-localization.

Cross-domain retrieval refers to the alignment and matching of candidate images from different sensors or imaging conditions. For example, Yan et al. [39] proposed an image reconstruction and multiunit mutual learning net (IML-Net), which constructs cross-domain feature descriptors using an image reconstruction network and a multiunit mutual learning module, and validated it on the multidomain dataset ZHcity750. Qiu et al. [40] proposed a method for selecting aerial

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images based on image moment extraction (IME), improving localization accuracy by optimizing invariant moment relative errors and area relative errors. Li et al. [41] proposed a cross-view localization framework based on patch similarity self-knowledge distillation (PaSS-KD), which jointly optimizes image restoration and localization tasks. By utilizing multiscale patch representations and the self-knowledge distillation mechanism, it effectively improves UAV localization accuracy under various weather conditions. Yang et al. [42] proposed a bidirectional visual geo-localization (VG) method that uses a cross-view matching network (CVM-Net) for cross-domain matching between digital twin images and the real-world images, achieving high recall rates on the Berlin dataset. Li et al. [43] proposed a multiscale attention encoder (MSAE) that integrates inverse polar transformation with a global mining strategy, thereby substantially enhancing image geo-localization accuracy.

Cross-modal matching refers to the alignment and matching of files with different contents or types. Map or road-network-assisted geo-localization methods align object features in images with map or road network information, combined with pose optimization, enabling efficient localization even in GPS-denied environments [6], [29], [44]. Samano et al. [29] proposed an embedding space-based geo-localization method, which improves image-to-map matching accuracy by learning embedding representations of panoramic images and map tiles. These approaches, falling under the umbrella of cross-modal geo-localization, inspire the possibility of using cross-modal matching-based retrieval algorithms for geo-localization.

B. Cross-Modal Retrieval With Remote Sensing Images

Many of the previously mentioned geo-localization methods are realized though retrieval algorithm with reference data. In recent years, cross-modal retrieval of remote sensing images has become a research hotspot, including sketch-to-remote-sensing-image retrieval [45], [46], [47], text-to-remote-sensing-image retrieval [48], [49], and video-to-remote-sensing-image retrieval [50], [51]. Based on the level of granularity, sketch-based retrieval can be categorized into category-level retrieval [52] and fine-grained retrieval [53]. Category-level retrieval focuses on identifying the category of a sketch, such as "bridge" or "river," to match corresponding remote sensing images, while fine-grained retrieval emphasizes the details and structure of the sketch to match specific instances. For category-level retrieval, Jiang et al. [45] proposed a multiscale network to address the problem of sketch-based aerial image retrieval. Xu et al. [47] developed a sketch-based remote sensing image retrieval (SBRSIR) method based on adversarial feature learning, which leverages discriminative loss to learn a deep joint embedding space and has been evaluated on sketch–remote sensing image datasets. Fang et al. [54] proposed a hashing-based remote sensing image retrieval method for scene localization using sketch queries. The approach constructs cohesive hash codebooks for sketches and remote sensing images, incorporating an interclass pairwise disperse cue (InterPDC) and an intraclass pairwise balance cue (IntraPBC) to enhance the discriminability and cohesiveness of the hash codes. With advancements in deep learning, attention mechanisms have been widely applied in sketch-to-remote-sensing-image retrieval, mitigating the modality gap between sketches and remote sensing images and enhancing retrieval accuracy [46], [55], [56]. To address the issue of category generalization in sketch retrieval, especially in applications involving remote sensing image datasets, zero-shot learning methods are becoming increasingly prominent [52], [56], [57], [58]. Chaudhuri et al. [57] proposed a cross-modal remote sensing image retrieval method based on label-deficit zero-shot learning (LDZSL). By training a deep learning model with a small amount of labeled data and aligning semantic information with visual features of remote sensing data, this approach overcomes the domain gap between different sensors (i.e., synthetic aperture radar (SAR) and multispectral), thereby enabling efficient retrieval of unseen class images.

Despite rapid progress in category-level SBRSIR, there remains a lack of development in fine-grained retrieval algorithms with cross-modal capability, thus preventing direct application to geo-localization. In our previous study [56], the proposed category-level retrieval algorithm demonstrated the potential for fine-grained retrieval, offering valuable insights that could inform future advancements in cross-modal fine-grained retrieval algorithms suitable for geo-localization.

III. Methodology

The proposed method introduces an innovative network architecture tailored for fine-grained cross-modal retrieval and endowed with strong zero-shot learning capabilities for geo-localization. Fig. 1 demonstrates the general framework of the proposed method. In this method, we first designed and implemented a rule-based automated pipeline for map generation and tiling. To achieve effective retrieval, we designed a fine-grained cross-modal retrieval model DSA-MLM with two core modules: the dual-stage attention module and the deep correlation matching module (DCMM). The model structure is shown in Fig. 2. The dual-stage attention module achieves efficient feature extraction and alignment through two stages: the adaptive feature extraction stage and the cross-modal alignment stage. Then, the DCMM performs fine-grained matching using the aligned features by assessing and ranking the similarity between the features of the query image and those of the map tiles. Based on the retrieval results, we employ a cluster-based geo-localization strategy to identify the optimal map tile group. Finally, the center point and its 50% confidence interval of the selected map tile group are calculated using Mahalanobis distance and define the location of the query image.

A. Cartography

We designed and implemented a rule-based automated pipeline for map generation and tiling. This algorithm reads a geo-spatial database and automatically generates multiscale map tiles containing linear symbols of different geographic objects with predefined scale parameters and drawing styles. The geographic features include roads, waterways, buildings, and land use information. The specific construction parameters and configurations are detailed in Table I. Subsequently, we

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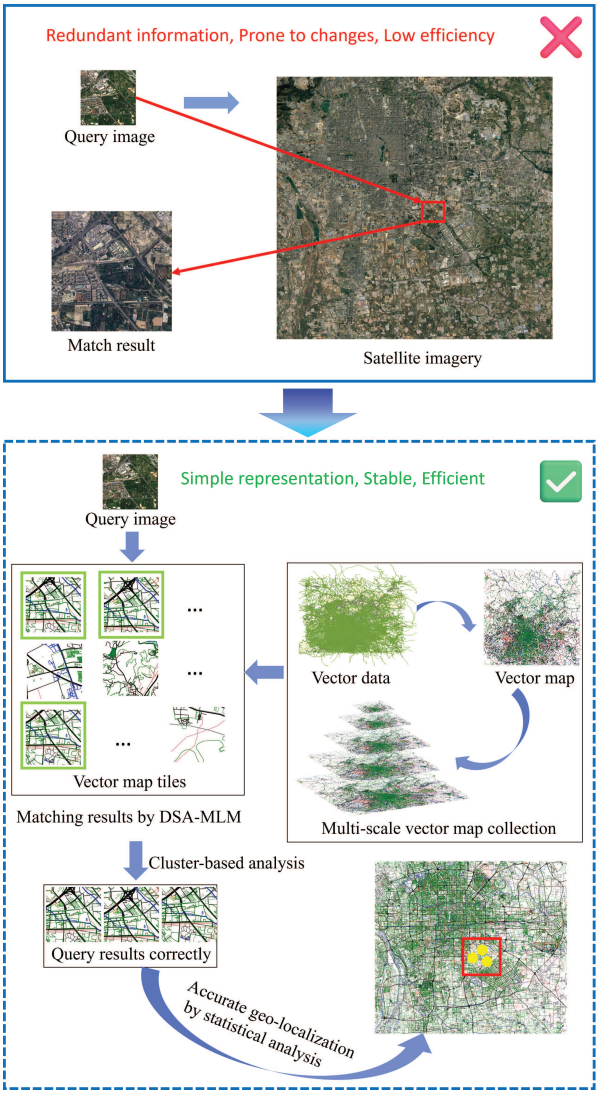


Fig. 1. Method overview: the solid line box represents the remote sensing image geo-localization using remote sensing image as a reference and the dashed line box illustrates the process with map tile database of linear symbol as a reference.

crop them into fixed-size map tiles. Each tile represents a well-referenced geographic bounding box. On this basis, we form a multiscale map tile, ultimately serving as the structured map tile database ready for retrieval. Limiting the map content to linear symbols markedly reduces file size. In a preliminary test, 200 tiles containing only linear features required just 72% of the storage space needed for an equivalent set of tiles that combined linear and surface features for the same region. Moreover, true overlap among linear symbols is uncommon. The linear features typically only touch or cross, so relying on linear data largely eliminates the occlusion issues that arise with overlapping surface symbols. Converting vector data into raster map tiles can introduce quantization errors that propagate to the final geo-localization results. To emulate practical deployment conditions, we rely on standard rasterization utilities and mitigate quantization errors solely through supersampling followed by downaveraging.

TABLE I
MAP CONSTRUCTION PARAMETERS AND SETTINGS

Type	Color	Specific Type Example	Line Width
Highway	Black	Motorway、Primary、Tertiary、Footway、Path Secondary、Path、Track	0.25-0.85
Waterway	Blue	Stream、River、Canal、Dam、Ditch、Drain	0.3-0.8
Barrier	Orange	Fence、Hedge、Wall	0.3-0.45
Railway	Gray	Abandoned、Disused、Construction、Platform、Rail、Station	0.2-0.45
Landuse Boundary	Green	Forest、Grass、Park、Orchard、Farmland	0.15-0.25
Building Boundary	Green	University、Industrial、Resi dential	0.2-0.3
Others	Red	Service、Pedestrain	0.2-0.4

B. Dual-Stage Attention Module

In the task of matching remote sensing images with map tiles, we propose an innovative dual-stage attention mechanism designed to enhance feature extraction and alignment between the two modalities. Unlike traditional multiscale attention mechanisms, this method enhances feature extraction and alignment between the two modalities through two key stages: intramodal adaptive feature extraction and intermodal cross-attention alignment. The core innovation lies in the dual-stage processing, which improves the precision of feature alignment, thereby significantly enhancing the effectiveness of remote sensing image and map tile matching, as detailed below.

1) Adaptive Feature Extraction Stage Within Modality: A map characterized by sparse linear symbols is not suitable for the ordinary Vision Transformer (ViT) [59]. ViT is a deep learning model that applies transformer architecture to image classification tasks by treating image patches as input tokens. In order to well capture map tiles' characteristics and overcome their linear symbols' inherent sparsity, we extend ViT with a multilevel feature extraction step. Specifically, both map and remote sensing images are progressively tiled into patches with different spatial resolutions to generate multi-scale fine-grained feature representations and finally coded into a high-dimensional feature space, respectively. During this process, convolutional layers and nonlinear activation functions are stacked layer by layer to increase the depth of the feature map, ensuring that subsequent network layers receive rich feature information from different levels. The feature extraction process at each layer can be represented as

$$X_{i+1} = f\left(\mathrm{Conv}_i\left(X_i\right)\right) \tag{1}$$

where X_i is the input feature map at layer i, Conv_i denotes the convolution operation for layer i, and f is the activation function. This multilevel feature extraction step allows the model to effectively capture multiscale information from the

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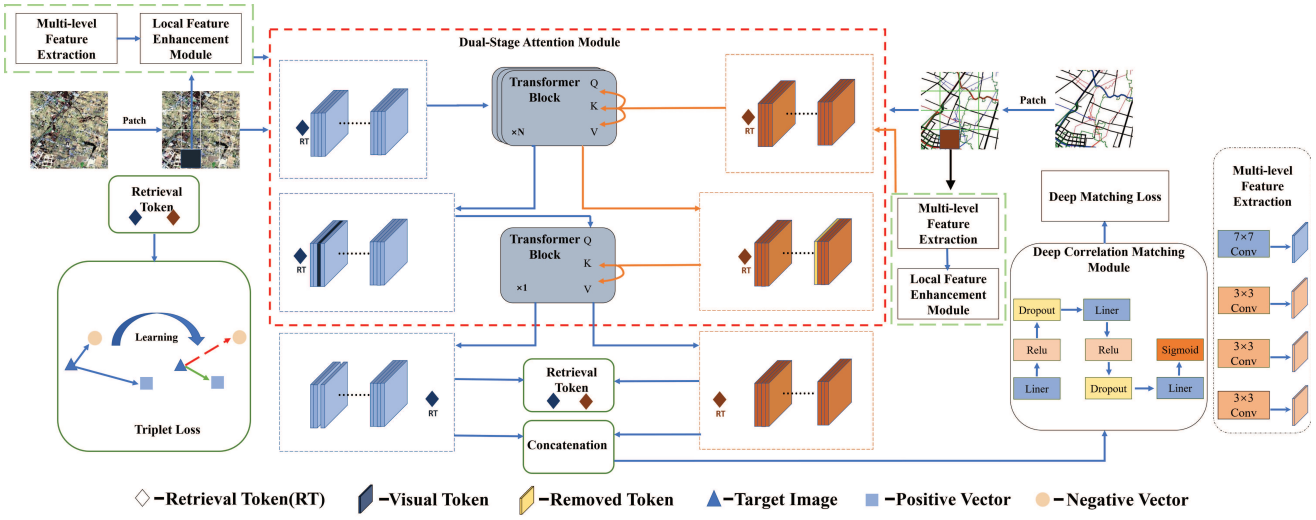


Fig. 2. DSA-MLM overview.

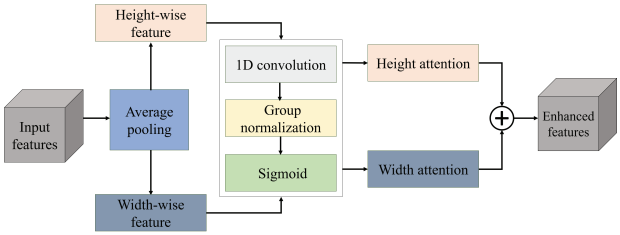


Fig. 3. LFEM overview.

image, providing a solid foundation for subsequent processing and analysis.

To improve the network's sensitivity to informative regions in both map tiles and images, we introduce a local feature enhancement module (LFEM) that applies spatial attention to the multilevel features extracted by successive convolutional layers (Fig. 3). The module first performs average pooling along the height and width of each feature map to obtain direction-specific statistics. These pooled features are then passed through a shared depth-wise separable 1-D convolution, followed by group normalization and a sigmoid activation, to generate spatial attention weights. The resulting attention maps for the height and width are fused with the original features via element-wise multiplication, thereby amplifying salient regions and enhancing the representational and discriminative capacity of the multilevel fine-grained features

$$A_d = \sigma\left(\mathrm{GN}\left(\mathrm{Conv1D}_d\left(\mathrm{AvgPool}_d\left(X\right)\right)\right)\right) \tag{2}$$

where d denotes the spatial dimension along which attention is computed, corresponding to the height or width direction; $\mathrm{AvgPool}_d$ represents average pooling along dimension d; $\mathrm{Conv1D}_d$ denotes the depth-wise separable 1-D convolution applied in the same direction; GN stands for group normalization; and σ denotes the sigmoid activation function.

We then rearrange the multiscale fine-grained features created in the multilevel feature extraction step into a sequential format. To preserve the spatial position information of each image patch within the original image, we introduce a positional encoding mechanism and integrate it with the feature representation of each patch through element-wise addition, thereby constructing an input representation with spatial awareness. Additionally, we prepend a global classification token (cls_token) to the feature sequence as an additional token for aggregating global information. This classification token is processed by the self-attention mechanism during the encoding process and ultimately serves as a comprehensive representation of the image's global features. To prevent the model from over-relying on specific patches or positional information, which could lead to overfitting, we employ a dropout strategy during training. By randomly masking certain patches or positional information, the model could learn more robust feature representations, thereby enhancing its generalization capability.

Subsequently, the feature sequences are fed into the transformer encoder. The encoder consists of multiple stacked encoding blocks, each containing a multihead self-attention (MSA) layer and a multilayer perceptron (MLP) layer. This structure captures the global dependencies and hierarchical information of each modality, enabling refined understanding. The input to each encoding block is first normalized using layer normalization to stabilize the data distribution and accelerate convergence. It is then passed through the MSA layer to calculate feature correlations between image patches. Specifically, the MSA maps the input features into query (Q), key (K), and value (V) matrices, computes the similarity between image patches, and generates an attention weight matrix. These weights are used to perform a weighted summation of the value matrix V, thereby producing a feature representation enriched with global contextual information. The scaled dot-product attention formula in MSA is as follows:

$$\mathrm{First_Attention}\left(Q, K, V\right) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{D}}\right)V \tag{3}$$

where D is the feature dimension of each token.

The output of the MSA layer undergoes dropout before being combined with the input via residual connections, preserving the original feature information while preventing overfitting. This refined representation is then passed through

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an MLP for further transformation. The MLP, which is composed of two linear layers and an activation function, projects the feature space to higher dimensional space, enabling the extraction of deeper features. The output of each encoder block is represented as

$$Z^{(l+1)} = \mathrm{MLP}\left(\mathrm{LN}\left(Z^{(l)} + \mathrm{MSA}\left(\mathrm{LN}\left(Z^{(l)}\right)\right)\right)\right) + \left(\mathrm{MSA}\left(\mathrm{LN}\left(Z^{(l)}\right)\right)\right) \quad (4)$$

where $Z^{(l+1)}$ is the input to the lth encoder block and LN denotes the layer normalization. This structure enables the encoder to progressively capture global associations in the imagery, providing enriched feature representations for subsequent tasks.

2) Cross-Modal Alignment Stage Between Modalities: Following the acquisition of features from remote sensing images and map tiles through the adaptive feature extraction stage, a cross-modal attention mechanism is introduced to establish the deep visual embedding association between the feature representations of the two modalities. The core of this method lies in exchanging the query representation of the two modalities (denoted as Q_I for remote sensing images and Q_V for map tiles) through a cross-attention mechanism. This enables interactive updates of both modalities' features across encoder layers, achieving tighter alignment within the cross-modal space. Specifically, after the exchange, the query, key, and value for the remote sensing images and map tiles are restructured into the forms (Q_I, K_V, V_V) and (Q_V, K_I, V_I), respectively. Through the cross-attention mechanism, the fusion of features from both modalities is achieved, ultimately resulting in the updated cross-modal feature representation. This cross-modal alignment stage follows the formula:

$$\mathrm{Second_Attention}\left(Q_I, K_V, V_V\right) = \mathrm{softmax}\left(\frac{Q_I K_V^T}{\sqrt{D}}\right) V_V. \quad (5)$$

Cross-modal alignment progressively integrates multidimensional visual features from remote sensing images and map tiles, generating updated cross-modal features that facilitate the similarity calculation between the two modalities.

Compared to traditional loss-based contrastive learning methods, the cross-modal cross-attention mechanism achieves a feedforward, feature-level alignment. The core advantage of this approach lies in its ability to proactively facilitate deep interaction between the features of the two modalities during the encoding process, rather than merely imposing passive constraints on sample distances at the loss calculation stage. Through the restructuring and exchange of *QKV*, the model can dynamically explore and integrate fine-grained correspondences between remote sensing images and map tiles, thereby generating more discriminative modality-invariant features.

C. Deep Correlation Matching Module

After completing the cross-modal alignment stage, we design the DCMM to achieve a deep similarity quantification of alignment between remote sensing images and map tiles in the feature space.

First, the features of remote sensing images and map tiles are partitioned by treating the first token as a global feature (cls_fea) and the remaining tokens as local features (token_fea); cls_fea represents the overall image feature, capturing global information, while token_fea contains spatially diverse local features with fine-grained details. To achieve effective dimensionality reduction while preserving spatial structure, token_fea is reshaped into a 3-D tensor and processed through a 2-D convolutional layer. Subsequently, through a flattening operation, the 2-D tensor of spatial features is transformed into 1-D feature vectors, resulting in a 1-D feature representation suitable for subsequent similarity computations.

Next, a similarity matrix is constructed by calculating the cosine similarity between the local features of remote sensing images and map tiles, in order to capture the degree of matching between the two modalities at the local feature level. The formula is as follows:

$$\mathrm{local_sim}(i, j) = \frac{\mathrm{im_fea}(i).vt_\mathrm{fea}(j)}{\|\mathrm{im_fea}(i)\|\|vt_\mathrm{fea}(j)\|} \quad (6)$$

where $\mathrm{im_fea}(i)$ and $vt_\mathrm{fea}(j)$ represent the ith and jth local feature points of a remote sensing image and map tile, respectively. Each element in the local similarity matrix reflects the similarity between remote sensing images and map tiles at local feature points.

To obtain a global similarity metric, the local similarity matrix is then input into a multilayer fully connected neural network, which progressively compresses the local similarity matrix into a single global matching score. The calculation process can be expressed as

$$\mathrm{dm_scores} = f_3\left(\mathrm{ReLU}\left(f_2\left(\mathrm{ReLU}\left(f_1\left(\mathrm{local_sim}\right)\right)\right)\right)\right) \quad (7)$$

where f_1–f_3 are linear transformations. The final scalar output is mapped to the range (0, 1) using a sigmoid activation function, yielding the final correlation score dm_score, which measures the overall matching degree between the remote sensing images and map tiles.

Compared with existing approaches to geo-localization and content-based remote sensing retrieval, our method exploits recent advances in multilevel feature retrieval and Vision Transformers—techniques that are still largely unexplored in both domains. To satisfy the unprecedented demands for fine-grained matching and zero-shot learning, we propose a novel architecture with two complementary attention-guided tokenization stages: 1) a within-modal attention module and 2) a cross-modal attention module. Together, these stages enable the network to capture subtle yet discriminative features both within and across the map and remote sensing modalities. To the best of our knowledge, such architecture is introduced here for the first time.

D. Losses

We employ the triplet loss and deep matching loss to enhance the distinction between positive and negative samples and to improve the accuracy of the correlation score prediction. The implementation and function of each loss function are detailed as follows.

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1) Triplet Loss: This loss function aims to minimize the distance between positive samples and anchor samples while maximizing the distance between negative samples and anchor samples. The objective is to ensure that the distance between a remote sensing image positive sample im_p and a map tile positive sample vt_p is less than the distance between the remote sensing image positive sample and a map tile negative sample vt_n. The triplet loss formula is as follows:

$$L_{\text{tri}} = \frac{1}{T}\sum_{i=1}^{T} \max\left\{d(\text{im}_p, vt_p) - d(\text{im}_p, vt_n) + m0\right\} \tag{8}$$

where d represents the Euclidean distance between remote sensing and map tile features, T is the total number of triplets, and m is the margin. The role of the margin is to force the model to separate the distance between positive and negative samples, ensuring that the model learns more discriminative features and avoids degenerate solutions.

2) Deep Matching Loss: Based on the mean squared error (MSE) loss, this loss function measures the accuracy of the global correlation score between remote sensing images and map tiles. Using the correlation scores generated by the DCMM as predictions, it calculates the squared difference between the prediction and the target. The formula is as follows:

$$L_{\text{dm}} = \frac{1}{N}\sum_{i=1}^{N} \left(\text{predict}_i - \text{target}_i\right)^2 \tag{9}$$

where N denotes the number of samples and predict_i and target_i represent the ith predicted and true values, respectively. To fully leverage the complementary effects of the two loss functions, the model's loss function L is the weighted summation of the triplet loss and the deep matching loss, defined as

$$L = \frac{1}{3}L_{\text{tri}} + \frac{2}{3}L_{\text{dm}}. \tag{10}$$

This weighting scheme aims to combine the ability to distinguish between positive and negative samples with the accuracy of correlation predictions. The triplet loss improves the embedding space by bringing similar samples closer together, thereby enhancing generalization. However, it does not directly optimize correlation prediction. The deep association loss focuses on improving predictions, which is why it is assigned a higher weight. However, it might overfit and not generalize as well as the triplet loss. By combining the two, the model ensures robust discriminative capability while enhancing correlation prediction accuracy, ultimately achieving superior cross-modal retrieval performance.

E. Cluster-Based Geo-Localization

Guided by the retrieval results, we apply the DBSCAN algorithm to spatially group the candidate map tiles. Each tile's centroid serves as a feature point. The neighborhood radius (ε) is set to the integer approximation of the circumscribed circle radius r of the query remote sensing image, and the minimum sample parameter is fixed to 1 so that every tile participates in clustering. The neighborhood radius ε of the DBSCAN algorithm is configured based on spatial geometric constraints to ensure accurate clustering of all candidate map tiles covering the same geographical region. The circumscribed circle radius r of the query remote sensing image provides a natural spatial scale metric for its geographical coverage. We set ε to an integer approximation of r, which establishes the core spatial constraint for the clustering process: it not only ensures that the algorithm operates within the appropriate spatial scope for effective search, thereby mitigating the fragmentation of correct matches due to an undersized radius, but also prevents the erroneous association of spatially irrelevant tiles caused by an excessively large radius. DBSCAN first computes pairwise Euclidean distances between tile centroids, flags adjacent tiles that satisfy $d(i, j) \leq r$ as core points, and then expands density-reachable links to form clusters. The resulting groups, therefore, strictly obey the spatial constraint, while isolated tiles are labeled as noise. The retrieval results are then sorted in descending order based on the group cardinality (tile number within a group), with the group of the highest cardinality prioritized as the optimal match. If multiple map groups share the same cardinality, the group with the lowest sum of ranking positions of its tiles in the retrieval results is selected as the optimal group. If the distances between the centroids of all tiles in the retrieval results fail to satisfy $d(i, j) \leq r$, the original ranking is preserved and the top-ranked tile is chosen as the best match with localization accomplished using its geographic coordinates. The detailed group selecting workflow is shown in Fig. 4.

Upon identifying the optimal map group, we performed normality tests on the spatial distribution of tile centroids within the group using the χ^2 QQ plot for Mahalanobis distance distribution (as shown in Fig. 5). In the QQ plot, the majority of data points closely align with the y = x reference line, indicating that the sample data conform to a bivariate normal distribution. However, a few data points at the tails show slight deviations, suggesting that the data exhibit a small degree of dispersion, which may reflect subtle differences between the actual data and the ideal bivariate normal distribution. Nevertheless, the overall data distribution can still be reasonably approximated as a bivariate normal distribution. The 50% confidence interval is calculated to

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2\left(1-\rho^2\right)}\left[\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{\left(y-\mu_y\right)^2}{\sigma_y^2} - 2\rho\frac{(x-\mu_x)\left(y-\mu_y\right)}{\sigma_x\sigma_y}\right]\right) \tag{11}$$

$$D^2 = \frac{1}{1-\rho^2}\left(\frac{(x-\mu_x)^2}{\sigma_x^2} - \frac{2\rho\left(x-\mu_x\right)\left(y-\mu_y\right)}{\sigma_x\sigma_y} + \frac{\left(y-\mu_y\right)^2}{\sigma_y^2}\right) \tag{12}$$

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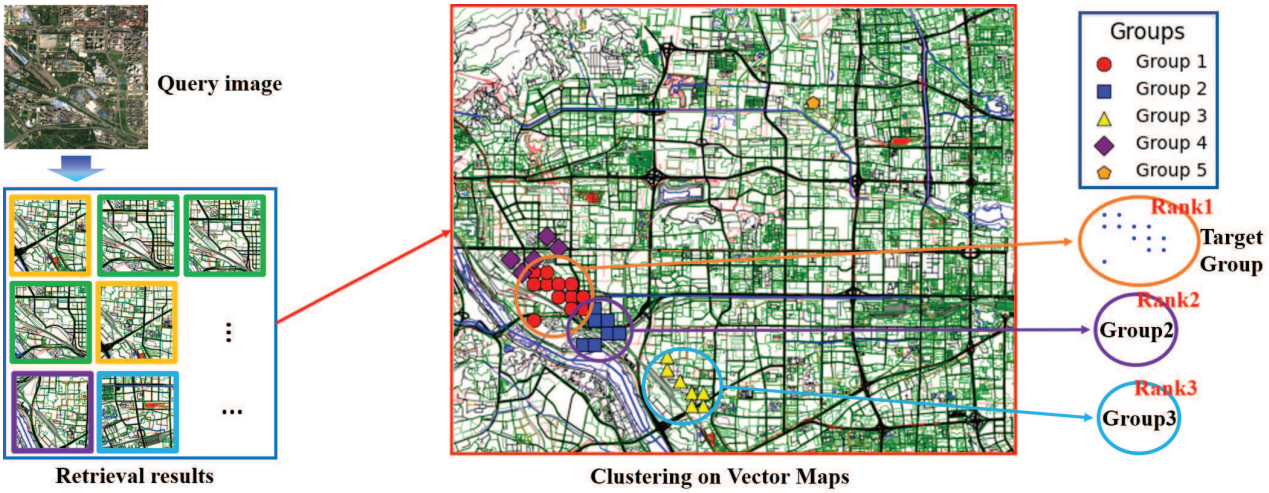


Fig. 4. Workflow of map tile clustering and geo-localization process: illustrating the steps of clustering map tiles based on distance constraints, ranking groups by cardinality, and selecting the optimal map group for accurate geo-localization (blue in this example).

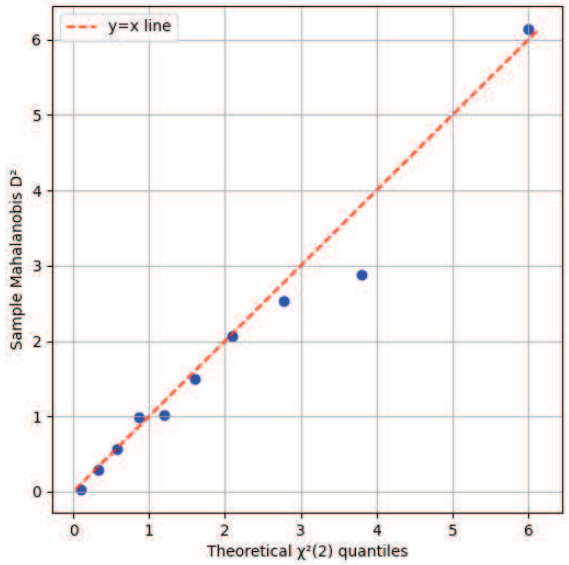


Fig. 5. χ^2 QQ plot for Mahalanobis distance distribution.

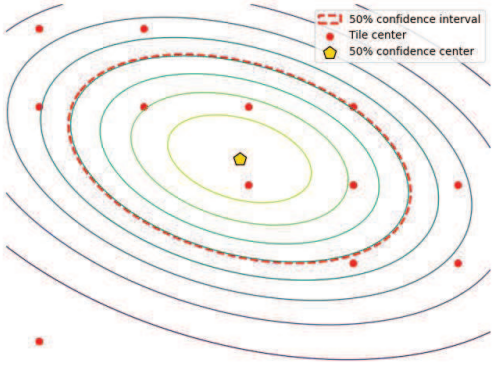


Fig. 6. Statistical distribution of map tiles within optimal map group.

define the geographic range of the query image. Subsequently, the Mahalanobis distance D between the center point of the query remote sensing image and the 50% confidence interval is calculated to determine whether the center point lies within the confidence interval. The bivariate normal distribution is illustrated in Fig. 6, with the calculation formula provided as in (11) and (12), shown at the bottom of the previous page, where x and y represent the coordinates of the center points of the map tiles; μ_x and μ_y denote the means corresponding to the x-axis and y-axis, respectively; σ_x and σ_y represent the standard deviations in the x-axis and y-axis, respectively; and ρ is the correlation coefficient between x and y, with a range of $-1 \leq \rho \leq 1$.

IV. Experiments

A. Experimental Setup

1) Datasets: We have constructed a new fine-grained remote sensing image retrieval dataset, FG_RSRD, for model training set and the testing set.

The training set consists of a total of 8061 image tiles, 7948 optical images from Sentinel-2, Landsat-8, Gaofen-1, and Gaofen-6, and 113 UAV SAR images, along with their corresponding map tiles. The dataset covers several urban, suburban, and rural areas in the middle and lower reaches of the Yangtze River in China, including Nanjing, Wuhan, Hefei, Anqing, and Chizhou. The dataset covers all four seasons. In the training set, optical satellite images of Sentinel-2, Landsat-8, Gaofen-1, and Gaofen-6 are cropped using the bounding box of map tiles. Each cropped image is then paired with its corresponding map tiles to form the training set, as shown in Fig. 7. The map tiles are created using OSM data. No additional offset has been added to linear symbols in the cartography process. Due to the crowdsourced nature of OSM, the map has variation accuracy and convergence in different regions, making it ideal to simulate the real-world conditions.

The testing set is divided into two subsets based on the spatial correspondence between remote sensing imagery and map tiles: FG_RSRD_RET (fine-grained retrieval testing set), in which the remote sensing image has the exact geographic bounding box of the corresponding map tile, and FG_RSRD_GEO (geo-localization testing set), in which the remote sensing image has an offset from its corresponding map tile, as shown in Fig. 8. The FG_RSRD_RET testing set have

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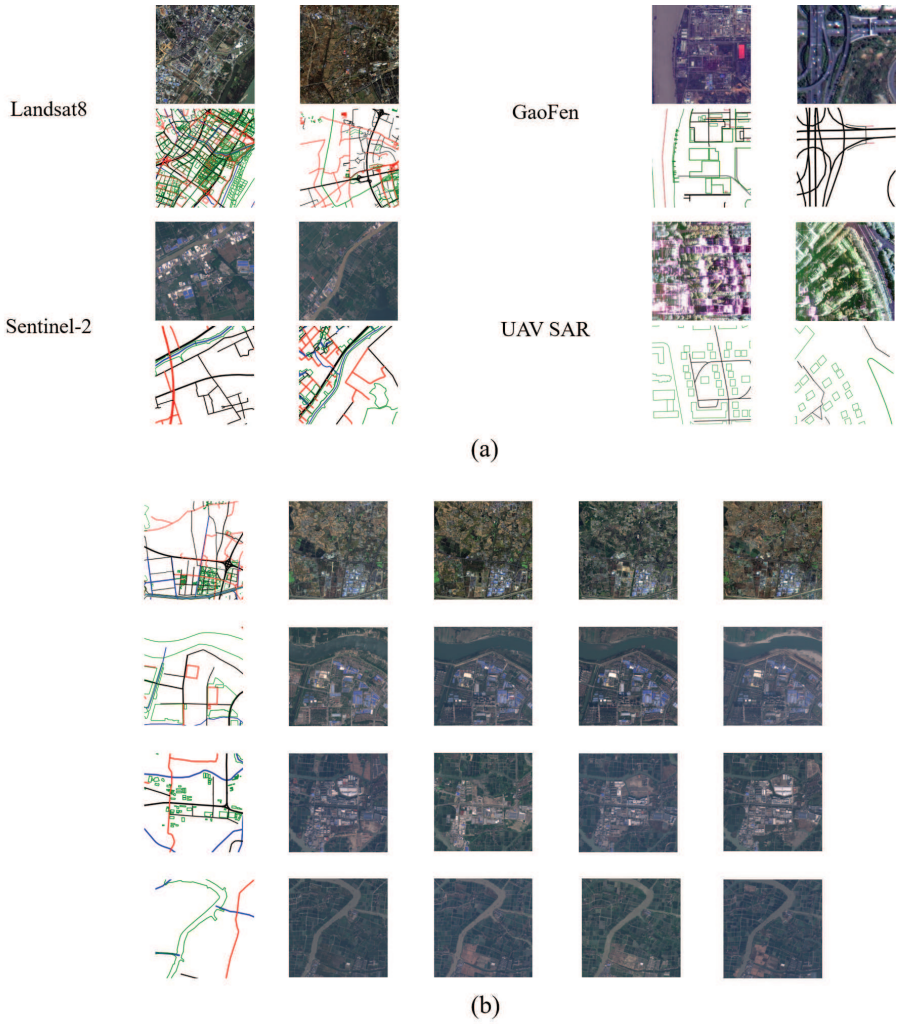


Fig. 7. Training set examples showing. (a) Remote sensing images from Landsat-8, Sentinel-2, Gaofen satellite, and UAV SAR, with their corresponding map tile data. (b) Map tiles and the four remote sensing images of the same area across different time periods.

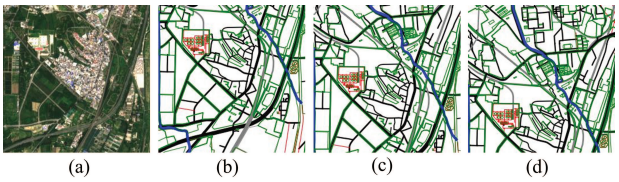


Fig. 8. (a) Sentinel-2 query image. (b) Map tile that completely corresponds to the query image. (c) and (d) Map tiles that are partially overlapping with the query image.

data in 1200 different geographic locations outside the training set, with each set consisting of four remote sensing images and one map tile. It is designed to evaluate the model's fine-grained retrieval performance under the condition of complete geographical match between remote sensing imagery and map tiles.

The FG_RSRD_GEO testing set, on the other hand, differs from those in the training set in terms of geographical region and data source in order to effectively test the model's performance and generalization capability. The remote sensing image in FG_RSRD_GEO has six resolution levels for optical images: 1, 10, 15, 20, 30, and 60 m, with a total of 1660 test images. Notably, the 15- and 30-m resolution remote sensing images were obtained by resampling the 10-m resolution images, covering areas ranging from 372.9 to 32 501 km^2. Also, the corresponding map tiles have multiple scales from 1:800 to 1:60 000. Fixed-stride map tiling results in boundary misregistration with query remote sensing imagery, creating local spatial offsets. For UAV SAR, the FG_RSRD_GEO testing set comprises 20 test remote sensing images and 69 580 map tiles, covering an area of 178.2 km^2.

2) Evaluation Metrics: To assess the performance of the model in retrieval and geo-localization test, we use top-K accuracy as the evaluation metric [60], which checks whether the correct retrieval result appears among the top K returned images, where K is set to 1, 5, 10, and 50. For retrieval result evaluation, the Euclidean distance d between the centroid of the query remote sensing image and the center point of the retrieved map tile is used to compare with the integer approximation of the radius r of the minimum bounding circle encompassing the remote sensing image. Choosing the integer approximation of the radius of the minimum bounding circle as the evaluation criterion balances computational simplicity, comprehensive geometric coverage, and adaptability to partially overlapping scenarios. If $d < r$ is satisfied, the retrieval

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TABLE II
TEST RESULT WITH FG_RSRD_RET DATASET

Method	FG_RSRD_RET			
	Top-1	Top-5	Top-10	Top-50
TripLet-RL	0.1100	0.1625	0.2425	0.3950
StyleMeUp	0.1400	0.2425	0.3125	0.5450
XModalViT	0.1350	0.1850	0.2025	0.2260
ACNet	0.1750	0.2280	0.2390	0.2490
MLAGT	0.3348	0.5379	0.6210	0.8260
Ours	**0.9531**	**0.9663**	**0.9738**	**0.9856**

result is deemed successful. This criterion allows for the evaluation of the model's accuracy under conditions where the remote sensing image partially overlaps with the map.

For geo-localization result evaluation, if the retrieval results cannot be clustered, the aforementioned $d < r$ criterion is directly adopted as the standard for determining successful geo-localization. If the retrieval results can be clustered, the Mahalanobis distance is further employed to assess whether the centroid of the query image lies within the 50% confidence interval derived from the chosen group of retrieved map tiles. If the centroid resides within this confidence interval, the geo-localization is considered successful.

3) Implementation Details: All images and map tiles are resized to 224 × 224 pixels. The training of the self-attention feature extraction module starts from the pretrained weights of ViT-B/16 [59] (Vision Transformer), which generates 196 visual tokens for each image, with each token represented as a 768-D vector. We employed the AdamW optimizer [61], with a learning rate set to $1e^{-5}$ and a weight decay of $5e^{-3}$. The model was trained with a batch size of 16 for 40 epochs. No gradient clipping was applied as training remained stable throughout the process. The total loss function was a weighted sum of the triplet loss and the deep matching loss, with weighting factors of 1/3 and 2/3, respectively. The model was built using the Pytorch framework, and all experiments were conducted on a Windows operating system with hardware consisting of a single NVIDIA RTX 4080 GPU and 32-GB RAM.

B. Experiments on FG_RSRD_RET

To validate the effectiveness of our model in fine-grained retrieval tasks, we first conducted experiments on the FG_RSRD_RET dataset, where the remote sensing images share the same geographic bounding box with the corresponding map tiles. Testing the proposed method in a pure retrieval setting without an offset map tile target eliminates interference from geographical approximation and, more importantly, facilitates comparison with existing cross-modal retrieval methods including Triplet-RL [53], StyleMeUp [62], XModalViT [63], ACNet [64], from the field of computer vision, as well as MLAGT [56] from the field of remote sensing. Given the relatively limited number of cross-modal retrieval methods in the remote sensing domain, we expanded the comparison scope by incorporating several representative methods from the computer vision field. The detailed comparison results are presented in Table II.

The results demonstrate that the proposed method significantly outperforms the existing methods on all evaluation metrics, particularly in terms of Top-1 accuracy, representing a 61.83% improvement over MLAGT. Furthermore, the proposed method also excelled in the Top-5, Top-10, and Top-50 accuracy metrics, clearly surpassing the performance of other methods.

C. Experiments on FG_RSRD_GEO

1) Optical Imagery: We conducted experiments on the FG_RSRD_GEO dataset. The retrieval results on Sentinel-2 are shown in Table III. The sample retrieval results in the Beijing region are illustrated in Fig. 9. Our proposed method demonstrates outstanding fine-grained retrieval performance even when there is a geographic offset between reference map tiles and the query remote sensing image. In the evaluation using Sentinel-2 imagery with varying spatial resolutions, the model's Top-50 retrieval accuracy remains consistently above 0.92, reaching a maximum of 0.9750. The retrieval results on Gaofen-2 images have Top-1, Top-5, Top-10, and Top-50 accuracies of 0.6250, 0.7850, 0.8350, and 0.9100, respectively. Example Top-5 retrieval results are illustrated in Fig. 10. Although Gaofen-2 imagery exhibits limited large-scale geographic features (e.g., roads and water bodies) in its map tiles, necessitating greater dependence on building feature extraction, our method demonstrates robust retrieval performance with a Top-50 accuracy of 0.9100.

On the other hand, the accuracy and median error in meter of the geo-localization results are detailed in Table IV, demonstrating the precision of geo-localization across different resolutions and imagery types. By utilizing Hausdorff distance as a spatial evaluation metric, we further validate the effectiveness and robustness of our method in terms of spatial accuracy.This model maintains high precision across a large search area up to 32 501 km^2, fully proving its broad applicability and reliability in real-world geo-localization tasks. These results indicate that our proposed method not only adapts to different geographic regions but also effectively addresses the challenges posed by varying resolutions and scales. It is worth noting that although the proposed method is effective in every tested resolution and data, the best result is achieved with an image resolution of 15 m. Higher resolution, like 1 m in GaoFen-2 imagery, may lead to a significant decrease in accuracy and an increase in median error. We attribute this primarily to insufficient feature representation in the map data itself, with the FG_RSRD_GEO map scale being a secondary factor.

2) UAV SAR Imagery: The proposed method achieves excellent localization accuracy in experiments on SAR images, with Top-1, Top-5, Top-10, and Top-50 accuracies reaching 0.8000, 0.9500, 1.0000, and 1.0000, respectively. The initial resolution of the imagery was approximately 0.3 m. To ensure that each image tile encompasses sufficient ground object features, the resolution was adjusted to approximately 3 m. The Top-5 retrieval results are illustrated in Fig. 11. Similarly, we also conducted a geo-localization experiment based on clustering, achieving an accuracy of 0.9000 within a distance deviation range of 0–150 m, further validating the effectiveness

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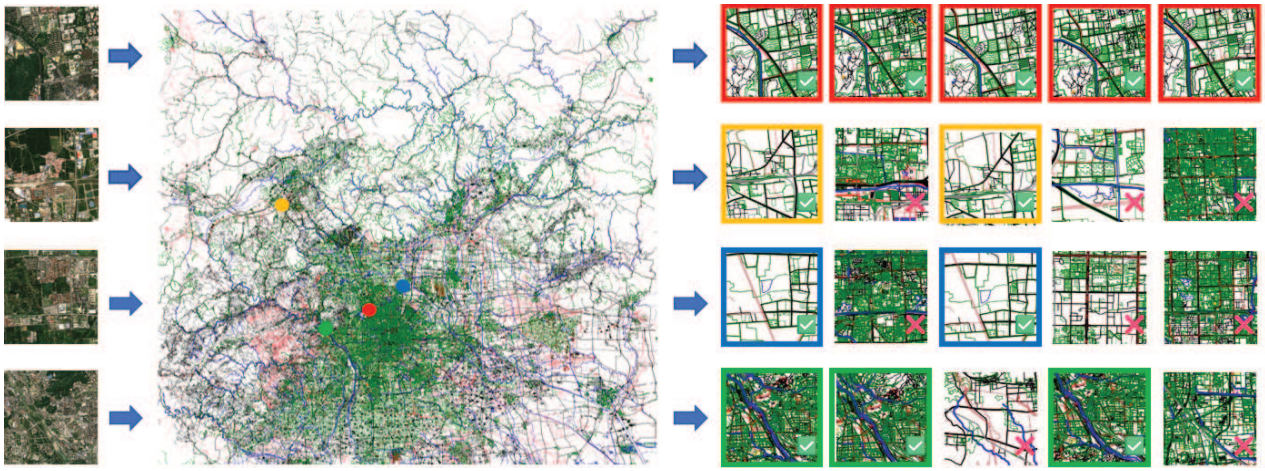


Fig. 9. Top-5 geo-localization results of Sentinel-2 imagery for the Beijing region on the FG_RSRD_GEO dataset (green checkmarks indicate correct answers, four different colored borders represent clustering results, and correspondingly colored dots on the map indicate geo-localization results).

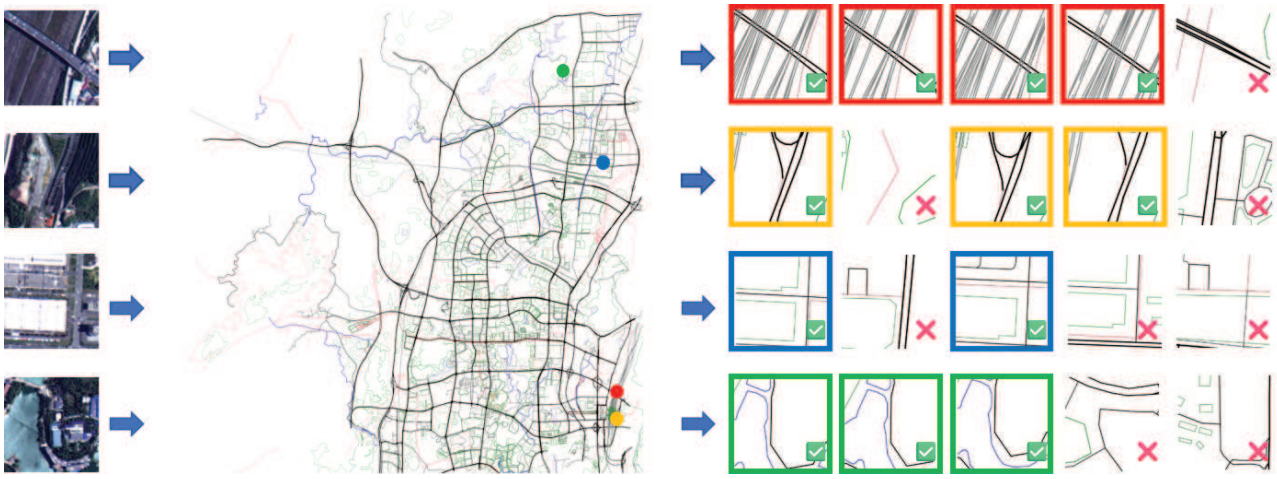


Fig. 10. Top-5 geo-localization results on Gaofen-2 imagery (green checkmarks indicate correct answers, red crosses indicate incorrect ones, four different colored borders represent clustering results, and correspondingly colored dots on the map indicate geo-localization results).

TABLE III
RETRIEVAL RESULTS ON THE FG_RSRD_GEO DATASET USING SENTINEL-2 IMAGERY

Resolution (m)	Beijing				Guangzhou			
	Top-1	Top-5	Top-10	Top-50	Top-1	Top-5	Top-10	Top-50
10	0.8260	0.8800	0.8980	0.9260	0.9220	0.9520	0.9600	0.9740
15	0.9275	0.9500	0.9550	0.9750	0.8350	0.9050	0.9275	0.9525
20	0.8575	0.9000	0.9300	0.9650	0.8875	0.9325	0.9450	0.9675
30	0.7750	0.8550	0.8800	0.9500	0.7200	0.8150	0.9000	0.9600
60	0.9167	0.9167	0.9167	0.9667	0.8000	0.8167	0.8833	0.9500

and robustness of the method in complex remote sensing environments.

The retrieval accuracy across different imagery types as a function of K value is presented in Fig. 12. The proposed method demonstrates consistently excellent performance across all tested datasets, with accuracy steadily increasing as K grows. Particularly, the method achieves the highest accuracy at relatively small K values when using Sentinel-2 (10 m) imagery, highlighting its powerful adaptability and representational capability for medium-resolution images.

3) Effect of Feature-Density Disparities: We randomly selected 200 Sentinel-2 images (10)-m spatial resolution) from vector-rich areas (urban and suburban) and vector-sparse areas (rural) as query samples. In vector-rich areas, the Top-1, Top-10, and Top-50 retrieval accuracies reached 0.915, 0.950, and 0.975, respectively. By contrast, in vector-sparse areas, these values fell to 0.565, 0.660, and 0.775. Although sparse vector information noticeably degrades retrieval accuracy, the subsequent clustering step stabilizes the geo-localization results, keeping them practically usable despite larger positional errors. Nevertheless, a lower bound threshold exists. When a query region contains too few linear features, the method may fail to produce a correct geo-localization. The other tested methods failed in this evaluation, indicating that

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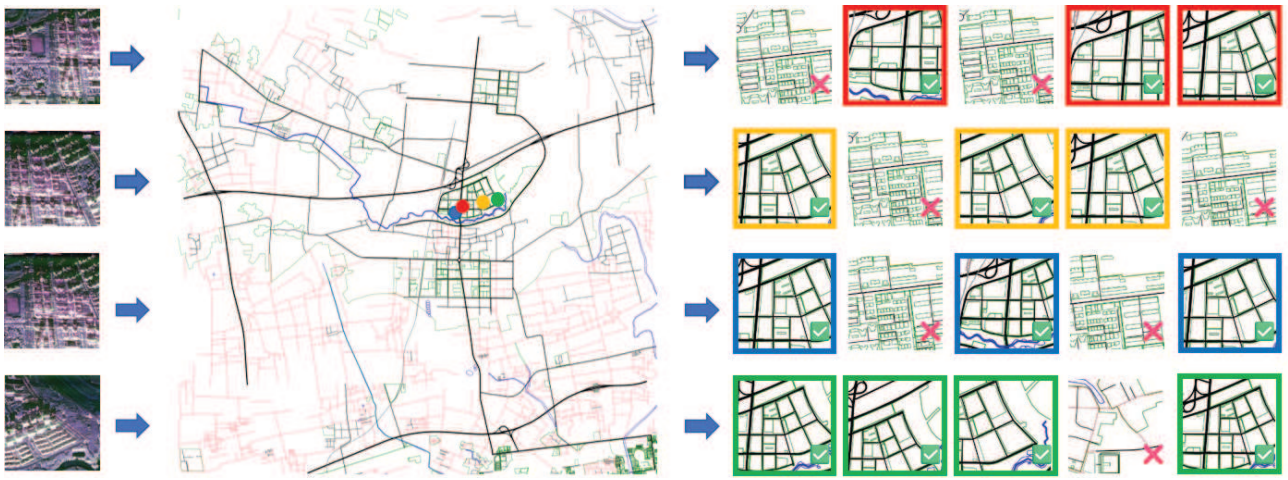


Fig. 11. Top-5 geo-localization results on SAR imagery (green checkmarks indicate correct answers, red crosses indicate incorrect ones, four different colored borders represent clustering results, and correspondingly colored dots on the map indicate geo-localization results).

TABLE IV
GEO-LOCALIZATION RESULTS ON THE FG_RSRD_GEO DATASET

Satellite	Resolution (m)	Accuracy	Hausdorff distance (m)	Median error (m)
GaoFen-2	1	0.7050	60.21	58.86
UAV SAR	3	0.9000	52.67	53.45
Sentinel-2	10	0.8640	104.43	102.50
Sentinel-2	15	0.9375	64.23	65.81
Sentinel-2	20	0.8700	233.54	222.52
Sentinel-2	30	0.8200	427.89	420.26
Sentinel-2	60	0.9167	613.27	616.80

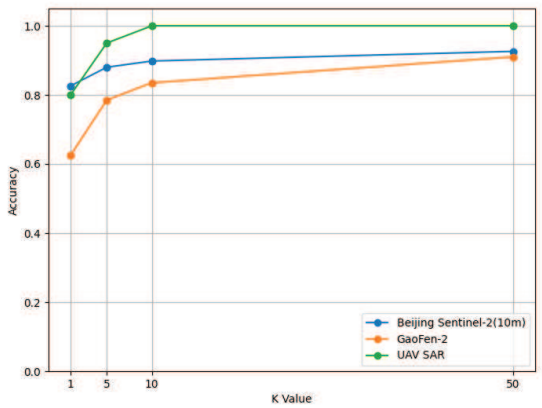


Fig. 12. Accuracy versus k-value of Beijing Sentinel-2(10 m), Gaofen-2, and UAV SAR.

they are considerably more sensitive to feature density than our proposed approach.

4) Computational Efficiency: Computational efficiency is mainly determined by the quantity of query images and reference maps in each experiment. In the experiment with Sentinel-2 optical imagery, the average query time per image was approximately 58 s, owing to the large coverage of the entire Beijing region (32 501 km^2). In comparison, for the entire Beijing region (32 501 km^2). In comparison,

TABLE V
PERFORMANCE COMPARISON BETWEEN LINEAR AND HYBRID SYMBOL MAPPING METHODS

Method	FG_RSRD_GEO		
	Beijing	Guangzhou	UAV-SAR
Conventional hybrid symbol maps	0.7683	0.8533	0.7500
Ours	**0.8260**	**0.9220**	**0.8000**

TABLE VI
MODULE ABLATION STUDY ON THE FG_RSRD_GEO DATASET

Components	10m	15m	20m	30m	60m
	Top-1	Top-1	Top-1	Top-1	Top-1
DSAN-FE	0.0000	0.0000	0.0000	0.0000	0.0167
DSAN-CM	0.8187	0.8250	0.7900	0.7833	**0.9167**
DCMM	0.2450	0.2483	0.3025	0.2000	0.1833
All	**0.8260**	**0.9275**	**0.8575**	**0.7750**	**0.9167**

for the smaller coverage area of the Guangzhou region (17 770 km^2), the average query time decreased to 42 s. For the UAV SAR imagery experiment, the map coverage area was significantly smaller than the Beijing region (178.2 versus 32 501 km^2), leading to substantially improved query efficiency, with average processing time reduced to approximately 14 s per image. Under these data volumes, memory usage ranged between 5.3 and 9.7 GB. Furthermore, a quantitative comparison was conducted between the proposed linear symbol mapping method and the conventional hybrid symbol mapping approach. Experiments were carried out using 10-m resolution Sentinel-2 imagery from Beijing and Guangzhou regions, as well as UAV SAR data from the Beijing area. The corresponding results are presented in Table V. The findings demonstrate that the linear symbol mapping method not only significantly outperforms the traditional hybrid method in terms of storage efficiency but also exhibits a notable improvement in retrieval and localization accuracy—achieving a consistent increase of

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TABLE VII
HYPERPARAMETER ABLATION STUDY ON THE FG_RSRD_GEO DATASET

Model parameters		10m Top-1	15m Top-1	20m Top-1	30m Top-1	60m Top-1
D_f	512	0.8060	0.8333	0.7125	0.7250	**0.9667**
	1024	**0.8260**	**0.9275**	**0.8575**	**0.7750**	0.9167
	2048	**0.8260**	0.7600	0.7050	0.7500	0.9500
Head	4	0.7800	0.8250	0.8333	0.7400	0.9000
	8	**0.8260**	0.9233	0.8400	**0.7750**	**0.9167**
	12	**0.8260**	**0.9275**	**0.8575**	**0.7750**	**0.9167**
	16	performance bottleneck				
Learning_Rate	1e-4	0.7740	0.8480	0.7950	0.7500	0.950
	5e-4	0.8200	0.8380	0.8400	0.7600	0.9333
	1e-5	0.8260	**0.9275**	**0.8575**	**0.7750**	0.9167
	5e-6	**0.8475**	0.8516	0.8467	0.7650	**0.9667**
	1e-6	0.8088	0.8250	0.8100	0.7400	0.9333
Weight_Decay	1e-1	0.7425	0.8380	0.7925	0.7100	0.9500
	5e-2	0.7850	0.8850	0.8275	0.7350	0.9667
	1e-2	**0.8260**	0.9233	0.8333	0.7650	0.9167
	5e-3	**0.8260**	**0.9275**	**0.8575**	**0.7750**	**0.9167**
	1e-3	0.8000	0.8683	0.7800	0.7700	0.9000

over 5% in Top-1 retrieval accuracy across different geographical regions and multisensor datasets.

5) Comparison With Related Works: Most existing studies rely on proprietary datasets for experimentation, which makes direct comparison between different algorithms difficult. Also, there are very few studies published with their code, making the algorithm reproduction difficult. Among the few studies from the computer vision field with publicly available implementations, we observed that their performance on our dataset was suboptimal, as demonstrated in Table II. To this end, we present relevant work in this section, aiming to provide valuable insights for methodological selection and advancement. Samano et al. [29] address the geo-localization problem by using a map as a reference for aircraft in scenarios where GPS signals are lost or unreliable. This study conducted experiments across multiple urban areas in the U.K. (ranging from 25 to 100 km^2), which are much smaller than our research area. In terms of retrieval accuracy, the Top-1 accuracy achieved 85.7%, 80.0%, and 75.0% in London, Oxford, and Bristol, respectively. The geo-localization results showed that the positional error typically ranged from 50 to 100 m. This method depends on an existing detailed map database that contains both linear and areal symbols. In contrast, our approach employs maps consisting exclusively of simple linear symbols, significantly reducing map file size and computational complexity for map generation. While Samano et al. [30] proposed semantic matching between panoramic images and 2-D maps for geo-localization, their validation was limited to small urban areas (2–3 km^2) with 0.72–0.82 Top-1 accuracy (>0.9 with turn information on 200-m routes). Our work significantly expands the scope to larger geographic regions, demonstrating superior generalization capability across more diverse and extensive environments.

TABLE VIII
CONFIDENCE INTERVAL ABLATION STUDY ON THE FG_RSRD_GEO DATASET

Confidence interval	Accuracy	Median error (m)
20%	0.8540	101.12
50%	0.8640	102.50
70%	0.8660	107.81
85%	0.8680	110.27
95%	0.8700	112.43

D. Ablation Study

We conduct several ablation studies based on the FG_RSRD_GEO dataset. Our goal is to comprehensively assess the performance of the method under different settings, thereby further optimizing the model's performance. To simplify the presentation, we report Top-1 accuracy in the following analysis.

1) Module: To validate the effectiveness of each module, we systematically remove individual components for experimentation. Specifically, in the dual-stage attention (DSAN) module, we first remove its feature extraction part (DSAN-FE) and replace it with ResNet-50 to extract features from remote sensing images and map tiles. Then, we retain the DSAN-FE module but remove the cross-modal alignment part (DSAN-CM) in DSAN, allowing the features extracted by DSAN-FE to directly enter the DCMM for matching. Finally, we remove the DCMM module and compute the similarity using only the feature vectors extracted by the dual-stage attention module. It is important to note that since the DCMM module is removed, the corresponding deep correlation loss can no longer be used. The Top-K retrieval accuracy in the module ablation study is shown in Table VI.

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TABLE IX
ABLATION ANALYSIS OF TILE STRIDE SIZE HYPERPARAMETER ON THE FG_RSRD_GEO DATASET

STRIDE	10m		15m		20m		30m		60m	
	Top-1	Top-50	Top-1	Top-50	Top-1	Top-50	Top-1	Top-50	Top-1	Top-50
1	0.0440	0.2840	0.0775	0.3475	0.1150	0.5625	0.0650	0.4850	0.0333	0.7000
1/2	0.0580	0.3440	0.4200	0.8250	0.1300	0.6275	0.0950	0.5950	0.2333	0.7667
1/4	0.4700	0.8040	0.3125	0.7375	0.2325	0.7025	**0.8100**	**0.9700**	0.2000	0.8000
1/6	0.5580	0.8260	0.5650	0.8650	0.6425	0.9000	0.5550	0.8700	0.5000	0.8333
1/8	**0.8260**	**0.9260**	**0.9275**	**0.9750**	**0.8575**	**0.9650**	0.7750	0.9500	**0.9167**	**0.9667**
1/10	0.8013	0.9038	0.8634	0.9550	0.8325	0.8738	0.8050	0.8750	0.7000	0.9333

The experimental results show that each module contributes to the fine-grained map tile retrieval performance of the model. Notably, when the feature extraction part of the model is removed and replaced with ResNet-50, the model fails to effectively perform feature extraction, making it difficult to support subsequent geo-localization tasks.

2) Hyperparameter: In this experiment, the following key hyperparameters were selected for ablation analysis. Learning rate influences the convergence speed and stability of the model during training.

Weight decay controls the regularization strength of the model to prevent overfitting.

The feature dimension in the cross-modal alignment module (D_f) determines the model's feature representation capacity.

The number of attention heads in the cross-modal alignment module influences the feature interaction capability of the cross-modal alignment module.

We set the baseline values as follows: the feature dimension in the cross-modal alignment module is 1024, the number of attention heads in the cross-modal alignment is 8, the learning rate is $1e^{-5}$, and the weight decay is $1e^{-3}$. All baseline parameters remain fixed, except for the ones being adjusted in the experiment. The specific experimental results are shown in Table VII.

The experimental results indicate that when the number of attention heads in the cross-modal alignment module is set to 16, it significantly affects the model's training speed, with the training time per batch being 8–10 times longer than for other configurations. When the feature dimension in the cross-modal alignment module is set to 1024, the number of attention heads is set to 12, the learning rate is $1e^{-5}$, and the weight decay is set to $5e^{-3}$, the model achieves the best fine-grained map tile retrieval accuracy.

3) Confidence Intervals: Following the clustering process, we set confidence intervals to define the spatial extent of geo-localization. Increasing the confidence level generally raises the proportion of successfully localized samples but also widens the interval, leading to higher median positional errors. Lower confidence levels have the opposite effect. They reduce errors but at the cost of fewer successful localizations. To quantify this tradeoff, we tested five confidence levels (20%, 50%, 70%, 85%, and 95%) on the FG_RSRD_GEO dataset using Sentinel-2 imagery at 10-m resolution. Table VIII summarizes the resulting localization accuracies and error statistics. The results indicate that a 50% confidence interval offers the best balance, yielding markedly lower errors than the 70% level while maintaining higher overall accuracy than the 20% level.

TABLE X
VECTOR COLOR ABLATION ON THE FG_RSRD_GEO DATASET

Method	10m	15m	20m	30m	60m
	Top-1	Top-1	Top-1	Top-1	Top-1
black	0.8060	0.8875	0.8125	0.1300	0.8667
colored	**0.8260**	**0.9275**	**0.8575**	**0.7750**	**0.9167**

4) Map Tile Stride: In practical applications, to improve the retrieval efficiency, it is necessary to minimize the number of tiles in the map, which can be achieved by increasing the stride of the tile slicing. However, this may degrade retrieval accuracy as the reference map tile may have a larger offset from the query image than when using smaller strides. Consequently, the cluster-based geo-localization yields coarser positioning results with increased median errors. Therefore, finding a reasonable balance between localization efficiency and retrieval accuracy becomes a key issue. To this end, we conducted experiments on the FG_RSRD_GEO dataset to explore the impact of different tile step sizes on geo-localization performance. In the experiments, the tile size was set to 256×256, and strides of 1, 1/2, 1/4, 1/6, 1/8, and 1/10 of the tile size were selected to crop map tiles at different scales. The results are shown in Table IX.

From the experimental results in Table IX, it can be observed that when the stride is set to 1/8 of the tile size, the model achieves the best retrieval accuracy. When the stride is too large, the generated map tiles fail to retain key feature information, resulting in a decline in retrieval accuracy. Conversely, when the stride is too small, although each tile can more accurately match the query image, the number of tiles in the map database increases drastically. This not only increases computational complexity but also amplifies interference, further impacting retrieval accuracy. Therefore, under the premise of balancing geo-localization accuracy and retrieval efficiency, we ultimately selected 1/8 of the tile size as the optimal stride size.

5) Mapping Rules: We investigate the potential impact of the visual variable color and thickness of line feature on retrieval accuracy during the mapping process. For color, the results presented in Table X indicate that when all linear symbols are uniformly represented in black, there is a significant fall in retrieval accuracy compared to the approach of using

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colors to distinguish feature types. We further examine how brightness affects retrieval performance by testing grayscale versions of the black map tiles at different brightness levels. The results show no significant difference in retrieval accuracy between grayscale and black map tiles. Additionally, we assessed the impact of color schemes on retrieval performance by testing different color combinations for linear symbol discrimination (e.g., roads, water bodies, and buildings). The experimental results demonstrate that these color scheme differences have a minimal impact on retrieval accuracy, with the variation being almost negligible.

Regarding the linewidth, we make specific allocations based on road categories (such as highways, primary roads, and secondary roads), as well as various land use types and building boundary types. Although we explored various combinations and incorporated random factors throughout the process, the findings indicate that these variations exert a negligible impact on retrieval accuracy, with no substantial alterations detected.

V. Conclusion

This article proposes a remote sensing image geo-localization method based on fine-grained retrieval of reference vector map tiles. The core of the method lies in the DSA-MLM retrieval model, which achieves cross-modal fine-grained retrieval, and the cluster-based geo-localization strategy, which is integrated with spatial constraint optimization to filter the optimal retrieved map group and determine the geographic boundary using the tile distribution within this group. Test results demonstrate that the proposed method achieves exceptional retrieval accuracy and geo-localization precision across multiple test scenarios. The method maintains stable high-precision performance under different but large geographic regions, image resolutions, and image sources, fully validating its robustness, generalization capabilities, and reliability in practical applications. Furthermore, since the map used in this method contains only linear symbols, it simplifies the reference map preparation.

Despite the achievements, the proposed method still requires further optimization in certain aspects. First, as the volume of remote sensing image data continues to grow, maintaining efficient processing capability on continental-scale datasets remains a pressing challenge. Future research may explore solutions combining updated deep learning and big data processing technologies to further enhance the efficiency and real-time performance of data processing. Second, the accuracy of the proposed method is influenced by the characteristics of reference map and its tile creation strategy, such as the scale of the map, the symbology rules, the feature density, and the tile stride. How to tailor and optimize reference map for a specific query dataset is a valuable future research direction. Third, using rasterized map tiles as the reference inevitably introduces quantization artifacts. Directly referencing the native vector database would eliminate this source of error, making it a promising avenue for future work. Finally, incomplete map coverage and positional inaccuracies pose practical constraints on real-world deployment. The OSM data used in this study exemplify these issues, showing uneven geographic completeness and, at times, outdated features. In regions with sparse or low-quality map content, the proposed method inevitably yields reduced retrieval accuracy—an inherent limitation of all map-referenced approaches. Developing strategies to mitigate this problem, such as incorporating additional data sources or enhancing robustness to sparse map content, represents a promising avenue for future research.

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